

ALVDALEN, Sweden

WMO: 023210

Lat: 61.2536N

Lon: 14.0355E

Elev: 253

StdP: 98.32

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-21.4	-26.5	0.3	-23.6	-23.8	0.5	-21.2	6.9	2.5	5.9	0.8	0.8	20	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	26.1	16.8	24.3	16.3	22.5	15.5	18.7	23.7	17.5	22.0	16.4	20.6	2.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.3	20.1	15.7	11.5	19.3	14.7	10.7	18.2	53.8	23.6	50.2	22.0	46.9	20.6	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.1	4.4	-27.3	28.2	2.5	1.4	-29.1	29.2	-30.6	30.0	-32.0	30.7	-33.8	31.7		
(5)			-27.5	19.8	2.3	1.2	-29.1	20.7	-30.5	21.4	-31.7	22.1	-33.4	23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-8.2	-6.6	-2.3	3.3	8.6	12.7	15.6	13.6	9.2	3.4	-1.7	-6.6		
	DBStd	9.25	6.48	5.57	4.87	3.39	3.78	3.07	2.83	2.74	3.09	3.85	4.59	6.45		
	HDD10.0	2820	564	464	381	203	72	9	1	4	49	206	352	516		
	HDD18.3	5436	823	697	639	452	301	171	94	149	273	463	602	774		
	CDD10.0	437	0	0	0	1	30	89	173	115	26	2	0	0		
	CDD18.3	11	0	0	0	0	0	1	8	1	0	0	0	0		
	CDH23.3	255	0	0	0	0	14	54	159	28	0	0	0	0		
	CDH26.7	33	0	0	0	0	0	5	27	1	0	0	0	0		
(14) Wind	WSAvg	1.6	1.3	1.4	1.8	1.8	2.0	2.1	1.8	1.6	1.6	1.5	1.4	1.3		
(15) Precipitation	PrecAvg	661	41	27	26	40	50	73	94	85	67	60	54	42		
	PrecMax	917	89	51	54	117	111	156	178	172	148	147	146	69		
	PrecMin	528	9	5	4	7	6	8	22	36	16	17	21	8		
	PrecStd	91	22	11	13	29	25	30	36	33	31	32	28	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	6.7	12.4	19.3	25.3	27.4	29.2	26.4	21.4	15.4	10.4	7.6		
		MCWB	3.6	3.6	7.0	10.6	14.4	16.0	18.9	17.7	14.2	11.9	9.3	5.5		
	2%	DB	4.0	4.5	9.4	16.6	22.7	25.0	27.2	23.8	19.0	13.0	8.2	5.3		
		MCWB	2.4	2.4	4.9	9.1	13.4	15.7	18.0	17.1	13.5	10.2	6.9	4.0		
	5%	DB	2.3	2.6	6.9	13.9	20.2	22.9	25.0	21.9	17.0	11.3	6.1	3.6		
		MCWB	1.1	1.1	3.2	7.5	12.1	14.9	17.2	16.2	13.0	9.1	5.1	2.5		
	10%	DB	0.6	1.1	4.9	11.4	17.1	20.3	23.0	19.9	15.2	9.8	4.5	1.7		
		MCWB	-0.2	0.0	1.9	5.9	11.0	13.5	16.5	15.2	11.9	8.2	3.7	1.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.1	7.0	11.6	15.6	18.2	20.1	19.4	15.8	12.6	9.4	6.0		
		MCDB	5.1	5.8	11.5	17.4	21.9	23.8	26.0	23.2	19.4	14.4	10.3	7.3		
	2%	WB	2.6	2.7	5.0	9.7	14.4	16.6	19.1	18.2	14.6	10.9	7.1	4.2		
		MCDB	3.7	4.3	8.9	15.9	20.9	22.2	24.9	22.0	17.7	12.5	7.9	5.0		
	5%	WB	1.3	1.3	3.5	7.9	12.9	15.5	18.1	17.0	13.4	9.6	5.3	2.5		
		MCDB	2.2	2.5	6.4	13.2	18.3	21.4	23.4	20.1	16.1	10.9	5.9	3.4		
	10%	WB	-0.1	0.0	2.2	6.4	11.5	14.3	17.0	15.9	12.2	8.3	3.7	1.0		
		MCDB	0.6	0.9	4.7	10.5	16.6	19.6	21.5	19.1	14.6	9.5	4.3	1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	9.2	11.0	11.9	11.8	11.5	11.5	11.0	10.4	8.3	6.1	7.1		
		MCDBR	7.7	9.9	13.0	17.7	18.2	16.6	15.8	14.6	13.5	10.6	7.8	8.4		
	5% WB	MCWBR	7.0	7.8	9.0	10.5	9.5	8.6	7.7	8.4	8.8	8.0	6.9	7.5		
		MCWBR	7.3	8.8	12.1	16.3	14.6	14.2	13.8	12.0	12.1	9.2	7.3	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.229	0.277	0.306	0.342	0.348	0.343	0.361	0.352	0.319	0.304	0.249	0.348		
		taud	2.162	2.279	2.317	2.372	2.444	2.485	2.447	2.464	2.541	2.485	2.243	2.775		
	Ebn at Noon		520	713	817	852	876	886	857	836	808	687	487	371		
		Edh at Noon	35	62	85	98	99	97	98	89	69	52	33	23		
(44) All-Sky Solar Radiation	RadAvg		0.28	0.95	2.52	4.03	4.70	5.29	4.81	3.71	2.43	1.10	0.34	0.15		
	RadStd		0.04	0.18	0.25	0.42	0.48	0.41	0.61	0.35	0.32	0.19	0.09	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page